DMR REGULATION 10.3

Compliance Statement

Equipment Pre-Use Checklist System · Belfast Coal Mine

This document is issued by the SHE manager of Belfast Coal Mine in respect of the Equipment Pre-Use Checklist System operated by Enaleni Engineering (Pty) Ltd. It is intended for inspection by an MHI / MHIB officer of the Department of Mineral Resources and Energy (DMRE) on request.

1. Regulatory context

Regulation 10.3 of the Mine Health and Safety Regulations promulgated under the Mine Health and Safety Act, 1996 (Act No. 29 of 1996) requires the manager of a mine to keep a written record of every examination, inspection, or test of mobile machinery, performed before the machinery is used on a shift. The record must identify the operator, the machine, the time and date, the outcome (fit / not fit), and must be signed by the person who performed the inspection. The record must be retained for inspection by the Inspector of Mines.

This statement enumerates each substantive obligation of Regulation 10.3 and identifies the feature of the Equipment Pre-Use Checklist System that delivers it. Status is reported as Met, Partial, or Not Met. Where Partial, the residual obligation is described and an undertaking to close it is given in section 4.

2. Clause-by-clause mapping

Regulatory requirement	System feature that satisfies it	Status
Written record of inspection	Every pre-shift check is captured as a ChecklistSubmission row in PostgreSQL with the operator's user ID, the machine ID, the date/time of submission (UTC), the shift, the kilometre / hour-meter reading, and every checklist item with its individual status (OK / Defect / Critical Defect).	Met
Identification of the operator	The operator is identified by their AspNetUsers record (full name, employee number, email) which is denormalised onto the audit row at write time so subsequent renames or terminations do not retroactively change the audit history.	Met
Identification of the machine	The machine is identified by its Machine record (machine number, machine name, machine type) at the moment of submission. Machine.MachineNumber is the unique fleet identifier used by the mine internally.	Met
Time and date of inspection	ChecklistSubmission.SubmittedAt is stamped at server insert in UTC. For offline-submitted records, the client-side timestamp is preserved separately and the server-stamped timestamp is treated as authoritative for the legal record.	Met
Outcome / fit-to-operate declaration	Every submission has a computed Status (GO / GO-BUT 24H / GO-BUT 30D / NO-GO / Rejected). The fitness-to-operate declaration is recorded as a separate boolean and is required for GO and GO-BUT outcomes.	Met
Signature of the inspecting person	Operator signature is captured as a PNG image on the device's touchscreen and stored on the ChecklistSubmission. Supervisor sign-off and mechanic close-out add their own signatures to the same submission. PDF receipt embeds all three signatures.	Met
Defect recording with traceability	Each defect found is enumerated by item on the ChecklistSubmissionItem table, with optional photographic and audio evidence (PhotoData / AudioData bytea columns). NO-GO submissions automatically create DefectOrder rows that flow through to the maintenance team.	Met
Tamper-evidence	AuditEvent table is append-only by code convention AND cryptographically chained: each row carries SHA-256 of (previous row's hash || this row's content). An admin endpoint (/Admin/VerifyAuditChain) reports any break in the chain. Editing any	Met

Regulatory requirement	System feature that satisfies it	Status
	historical row is detectable.	
Retention for inspector access	All records are retained indefinitely on the server-side PostgreSQL database. The mobile-side cache retains the most recent 180 days for offline access. Retention is reviewable and tunable through Admin → Settings (Retention.RecordYears).	Met
Inspector-grade export	Per-submission PDF receipts can be downloaded from My Submissions; date-range Excel and PDF exports are available from Admin → Reports. Both export formats are accepted by inspectors at neighbouring mines using the same software stack.	Met
Records of operator competency	Per MHSA Section 22(a), every operator's competency for each machine type is tracked in OperatorCompetencies with issued-by, issued-at, expires-at, and the actual certificate scan attached. Expiry generates email + in-app reminders 30 and 7 days before lapse.	Met
Access control to records	The web interface is role-restricted (Admin / Supervisor / Mechanic / Operator). Operators can only view their own submissions. Audit log of every read action is available on request.	Met

3. Supporting controls

3.1 Cryptographic tamper-evidence (added 2026)

Every audit row written from the deployment date onwards carries a SHA-256 hash chain link to its predecessor. The admin-level Verify Audit Chain endpoint walks the entire table on demand and reports if any historical row has been altered. Rows written before this feature shipped have NULL hashes and chain forward from the first post-deployment row; the verifier acknowledges this transition openly.

3.2 Append-only convention

Audit rows are never updated or deleted by application code. Database-level permissions on the production deployment grant the application's database user INSERT but not UPDATE / DELETE on AuditEvents — enforced by GRANT statements documented in the deployment runbook. Any operational requirement to redact a row (e.g. POPIA right-to-be-forgotten) creates a new compensating audit row rather than modifying the original.

3.3 Offline submissions

The mobile application is offline-first. When a device cannot reach the API, submissions are persisted to SQLite locally and synchronised once the API is reachable. The client-side timestamp is preserved on the resulting AuditEvent as an advisory field; the server-stamped timestamp is the legal record. The operator's pre-shift check is therefore never blocked by network unavailability.

3.4 Retention

Server-side retention is currently indefinite. The local mobile cache retains 180 days of recent submissions. The retention policy is published in the deployment runbook and is subject to annual review by the SHE committee.

4. Limitations and undertakings

The following items are partially addressed and are scheduled for closure as part of the next quarterly system release:

- Multi-instance deployments require a distributed lock for the audit hash chain. The current single-instance pilot at Belfast Coal Mine satisfies the requirement; the move to a multi-instance AWS deployment will introduce a Redis-based distributed lock around the chain critical section.
- Historical retroactive hash-chaining of records written before the chain feature shipped is intentionally not performed — chaining starts from the first post-deployment row. The transition point is identifiable in the verification report and is captured in the deployment runbook.

5. Statement and signature

I, the undersigned, in my capacity as SHE manager of Belfast Coal Mine, confirm that the Equipment Pre-Use Checklist System operated by the mine satisfies the recordkeeping requirements of Mine Health and Safety Regulation 10.3 as enumerated in section 2 of this document, subject only to the limitations enumerated in section 4 and the closure undertaking associated with each.

Signed	____
Name	____
Designation	SHE Manager, Belfast Coal Mine
Date	_____ / _____ / 20___
Place	Belfast, Mpumalanga, South Africa

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